The Invariant Content of Equivalent Theories

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spoiler alert!

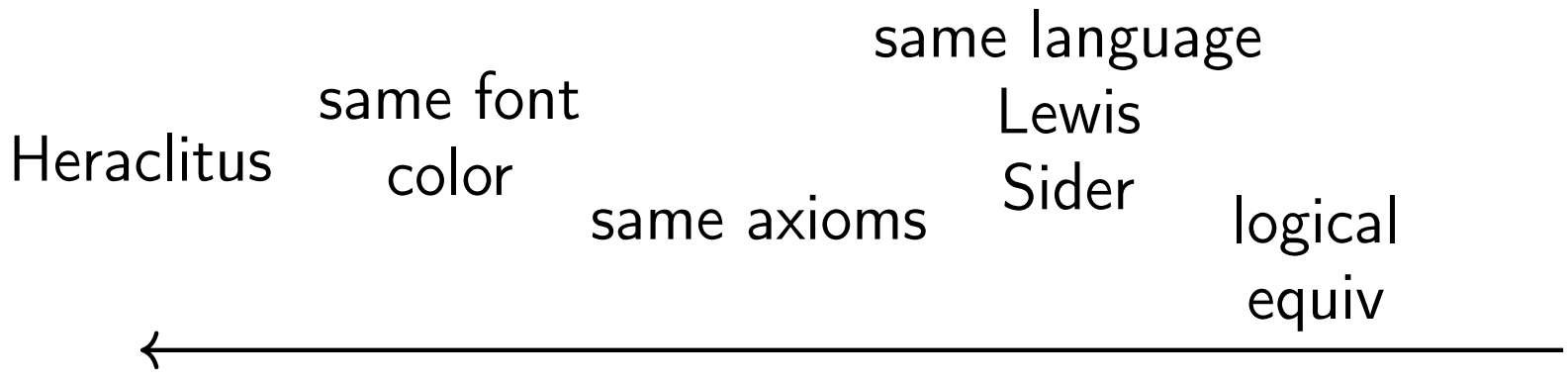
It isn't much!

It isn't nothing!

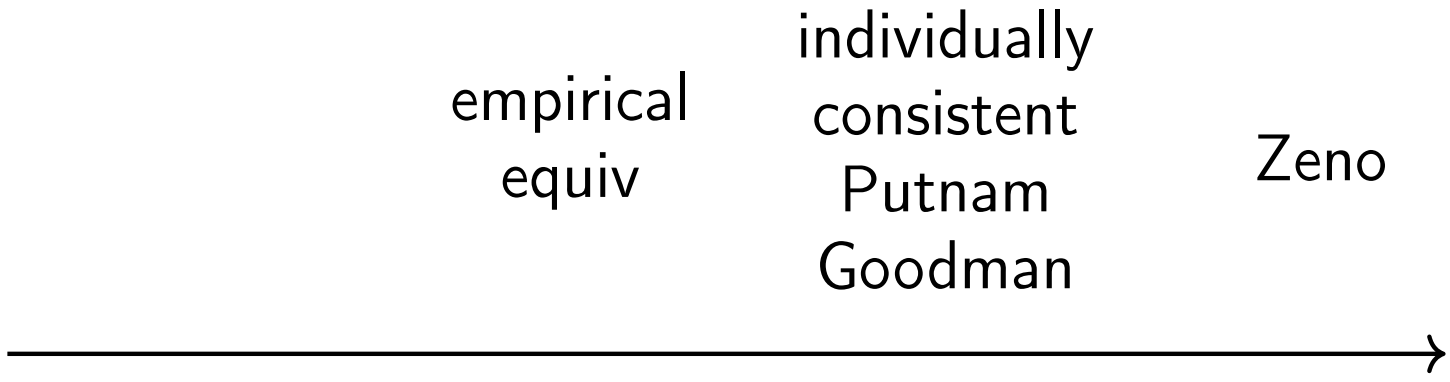
pragmatic prolegomena

- When can debate end?
- When is more investigation called for?
- For what should I be willing to take a stand?
- Invariants as necessary conditions

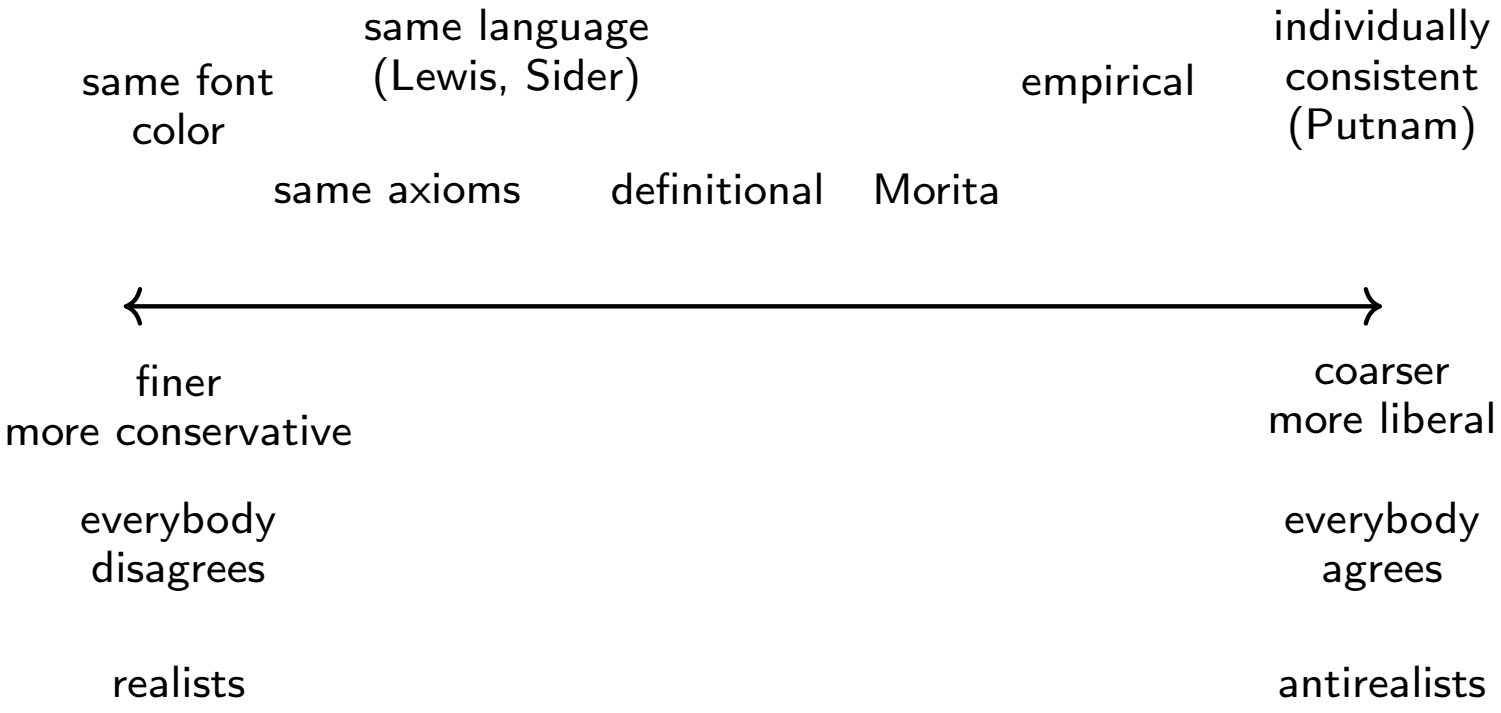
radical right wing



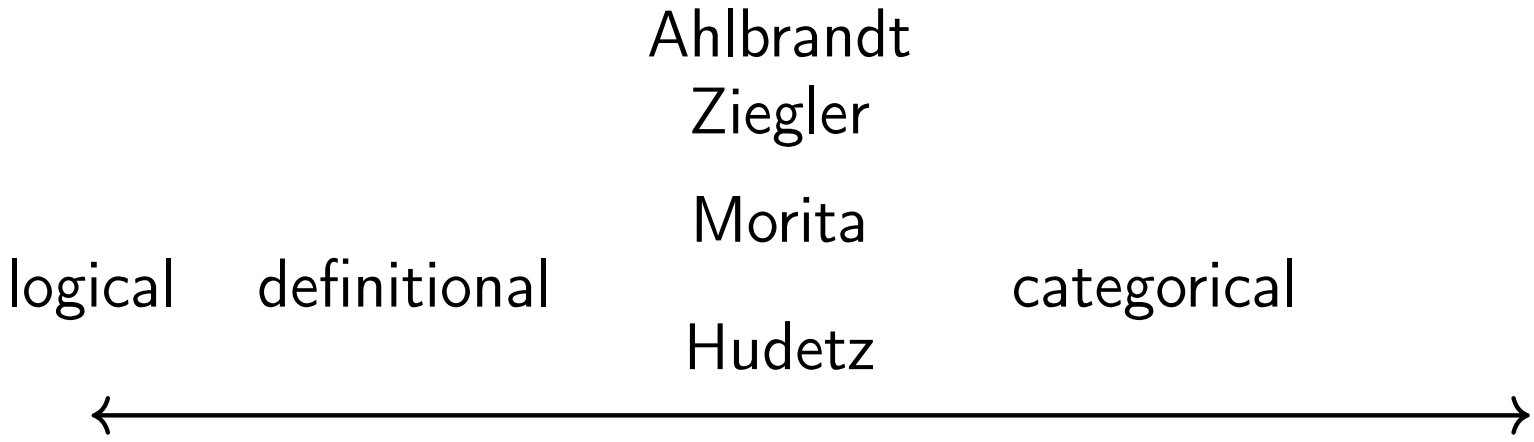
radical left wing



the ideological spectrum



moderate views



logical equivalence

- T_1 and T_2 are **logically equivalent** just in case they are in the **same** signature and:

$$\mathbf{Cn}(T_1) = \mathbf{Cn}(T_2)$$

$$\mathbf{Mod}(T_1) = \mathbf{Mod}(T_2)$$

- Overly restrictive: same signature

definitional extension

- Example: define $\leq$ by

$$\forall x, y(x \leq y \leftrightarrow (x < y \vee x = y))$$

- **Defn:** T^+ is a **definitional extension** of T just in case for each new relation symbol r of Σ^+, there is a formula Fr of Σ such that:

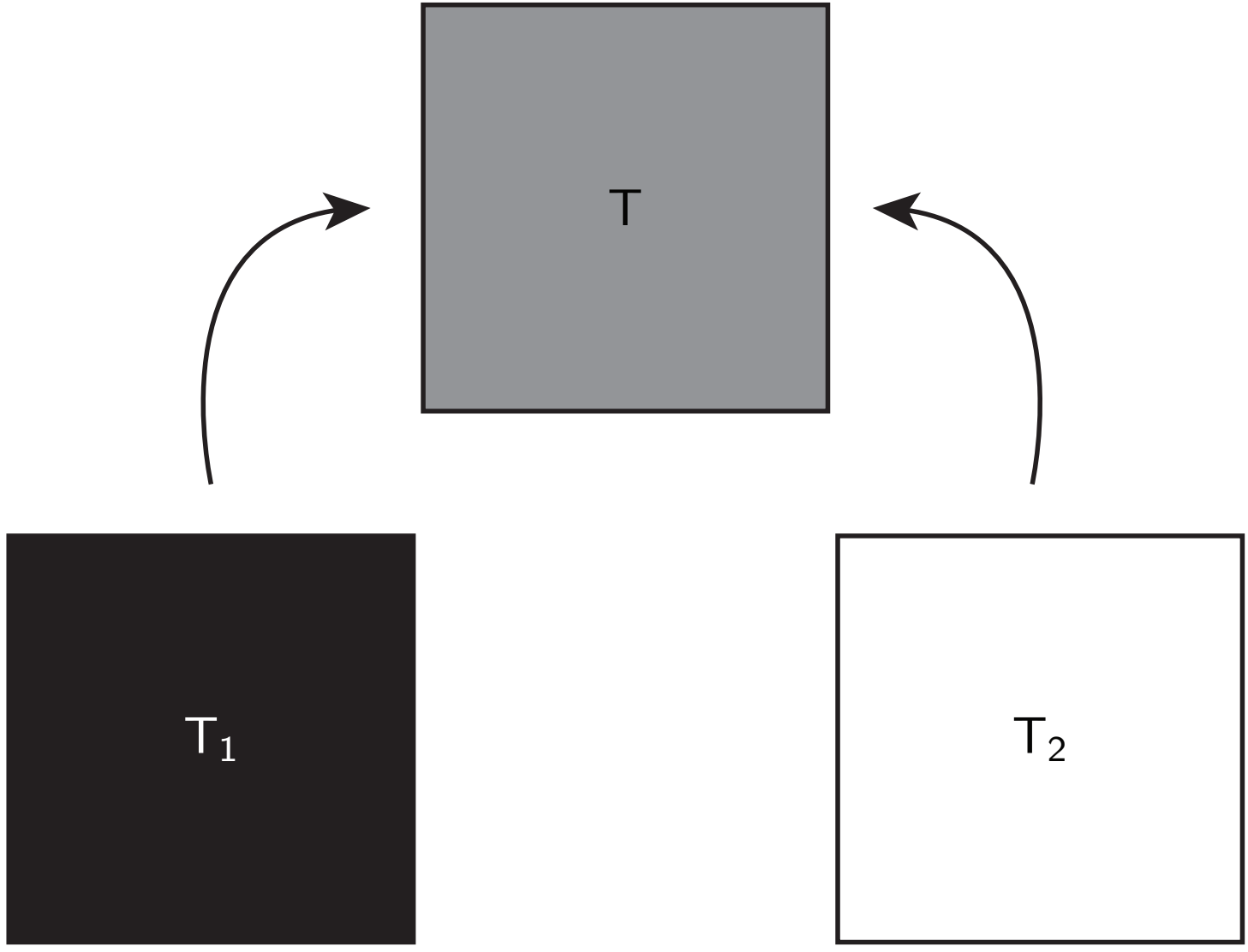
$$T^+ \vdash \forall \overline{x}(r(\overline{x}) \leftrightarrow (Fr)(\overline{x})).$$

definitional extension

- $r \mapsto Fr$ extends to an **interpretation** $F : T^+ \to T$.
- Each model M of T is uniquely expandable to a model $F^*(M)$ of T^+, where

$$M \vDash (F\phi)(\bar{a}) \quad \Longleftrightarrow \quad F^*(M) \vDash \phi(\bar{a})$$

T_1 and T_2 are **definitionally equivalent** if they have a **common definitional extension**.



definitional equivalence

Prop: T and T' are definitionally equivalent if and only if there are (one-dimensional) interpretations $F : T \to T'$ and $G : T' \to T$ that are quasi-inverse to each other.

invariants of DEF

1. numerical claims
2. cardinality of models
3. definable subsets of models

reasons to liberalize

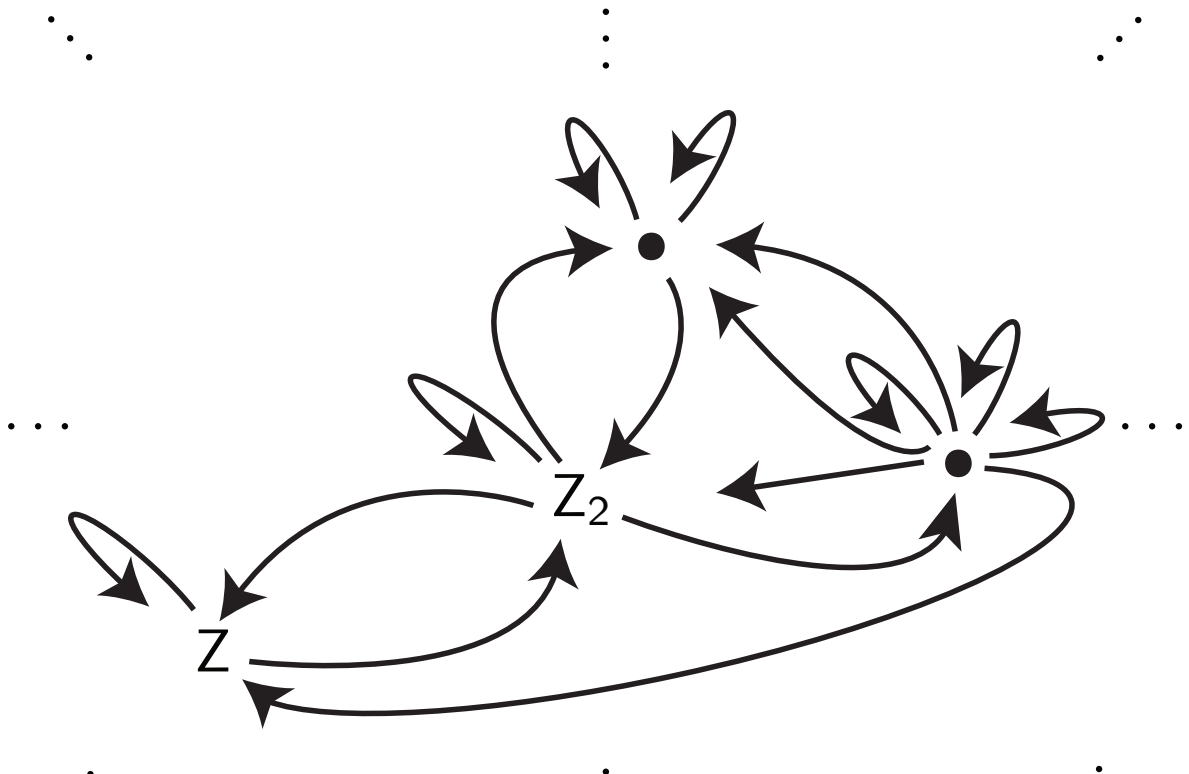
- Aim: give a rational reconstruction of actual scientific and mathematical practice.
  - Equivalence of wave and matrix mechanics
  - Tarski's axiomatizations of geometry
  - Dualities in math and physics
- Mathematics (and logic!) have their own internal standards of fruitfulness.

categorical equivalence

Consider $\mathbf{Mod}(T)$ as consisting of two kinds of things:

objects models of T

arrows elementary embeddings of models



categorical equivalence

objects of $\mathbf{C}$ $\longmapsto$ objects of $\mathbf{D}$

arrows of $\mathbf{C}$ $\longmapsto$ arrows of $\mathbf{D}$

(1) plays nicely with composition of arrows

(2) quasi-invertible

$$\text{logical} \implies \text{definitional} \implies \text{categorical}$$

Goodmania?

T_1 There is exactly one thing, named c_1.

T_{17} There are exactly seventeen things, named $c_1, c_2, \ldots, c_{17}$

 $\mathbf{Mod}(T_1) \equiv \mathbf{Mod}(T_{17})$ 

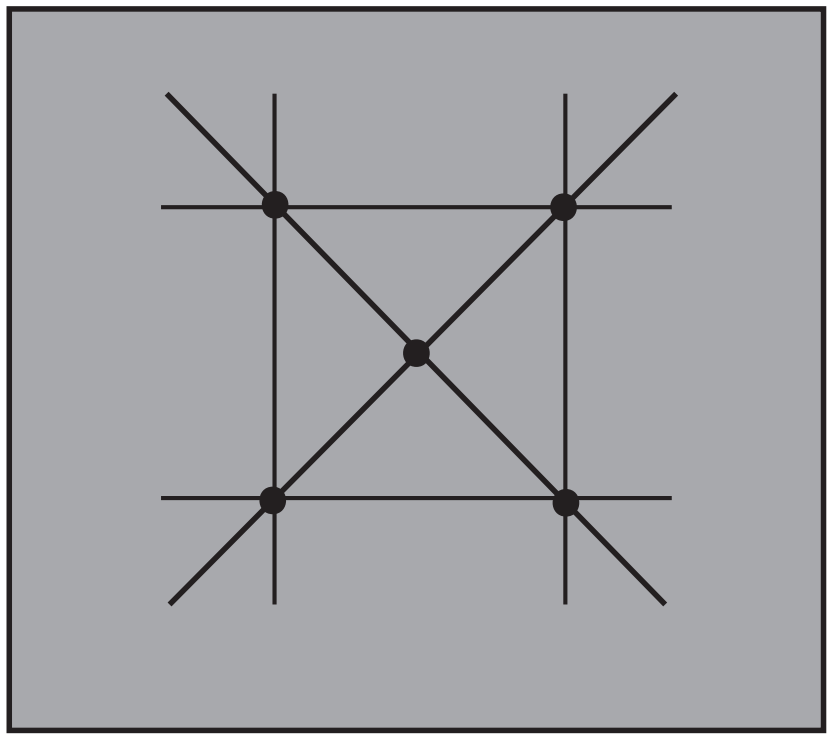
Morita equivalence

- **many-sorted logic:** Signatures include **sort symbols**, rather than just relation symbols.
- Morita extension:
  - product sorts
  - quotient sorts
  - subsorts
  - coproduct sorts
- **Defn:** T_1 and T_2 are **Morita equivalent** just in case they have a common Morita extension.

Morita equivalence

Morita $\Longrightarrow$ categorical

Invariant lost: cardinality of models



$$\text{logical} \implies \text{definitional} \implies \text{Morita} \implies \text{categorical}$$

Ahlbrandt-Ziegler equivalence

- An m-dimensional **interpretation** of T in T' assigns to each formula $\phi(x_1,\ldots,x_n)$ of Σ, a formula $\phi^*(\vec{y}_1,\ldots,\vec{y}_n)$ of Σ'.
- An interpretation of T into T' naturally induces a functor from $\mathbf{Mod}(T')$ to $\mathbf{Mod}(T)$.

Ahlbrandt-Ziegler equivalence

Defn: T and T' are **AZ equivalent** just in case there is an interpretation of T into T' that induces an equivalence between the categories $\mathbf{Mod}(T')$ and $\mathbf{Mod}(T)$.

$$\text{logical} \implies \text{definitional} \implies \text{AZ} \implies \text{categorical}$$

conjecture

Morita $\overset{?}{\Longleftrightarrow}$ AZ

application: Hamiltonian and Lagrangian mechanics

- J. North: "Hamiltonian and Lagrangian mechanics are not equivalent in terms of statespace structure. This means that they are not equivalent, period."
- T. Barrett: Hamiltonian and Lagrangian mechanics are categorically equivalent.

application: acceptance

- ACCEPT: To accept a theory T is to believe that some model of T is *isomorphic* to the world.
- EQUIV If T is equivalent to T', then you can't accept T without accepting T'.
- $\mathrm{MORITA} + \mathrm{EQUIV} \Longrightarrow \neg\mathrm{ACCEPT}$

application: wavefunction realism

- **Hilb** is categorically equivalent to **CStar**
- **CStar** doesn't talk about wavefunctions.
- Therefore, you should take your wavefunction *cum grano salis*.

conclusions

- Carnapian tolerance: there is no question of *correctness* when it comes to notions of theoretical equivalence.
- Second order morals: (1) clarity, (2) tolerance of dissenters.